

Baselines and fair comparisons

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Hypotheses and measurable outcomes

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Replication and reproduction

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reading a research claim

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Baselines and fair comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Baselines and fair comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Model uses 12 seconds and baseline 2 on each case. Task has a 3-second budget. How should results be framed?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Assistant gets current register; baseline gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A baseline is credible relative to the task, not merely a procedure easy to outperform. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Model uses 12 seconds and baseline 2 on each case. Task has a 3-second budget. How should results be framed?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled baselines and fair comparisons evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of baselines and fair comparisons. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

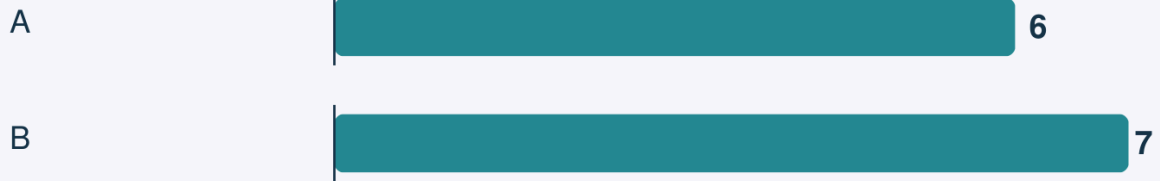
12. Pictorial case practice: Baselines and fair comparisons

A baseline deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

Simple rule A: 6/8 correct. New model B: 7/8 correct. No test case is used to tune either version.

Two versions on eight shared cases

correct answers out of 8



Bar length shows the supplied value; the marked baseline is zero.

Baselines and fair comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare baseline

A uses a simple documented rule rather than an intentionally weak placeholder.



2. Measure fairly

A is 75%; B is 87.5% on the same eight cases.



3. Interpret cautiously

B adds one correct case here; compare resources and failure types too.

CHANGE ONE THING

Give B the answer key before testing.

Explain the changed result using the matching lesson.

Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A baseline deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B adds one correct case here; compare resources and failure types too.

1. Declare baseline
2. Interpret cautiously
3. Measure fairly

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Interpret cautiously B adds one correct case here; compare resources and failure types too. A baseline anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

Practice 2: explain the deciding evidence

Changed-case practice: Give B the answer key before testing. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The resulting comparison is invalid as unseen evaluation.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The resulting comparison is invalid as unseen evaluation. B would receive privileged information that A did not receive.

Repair a missing building block

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the mechanism and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the mechanism and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the mechanism and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

